

Solturio

Decentralized Logo IP Protection Platform

Plant Your Standard on Chain - A CATH Ecosystem project

Overview

Solturio is a decentralized web application that enables creators, brands, and projects to protect their intellectual property by registering logos on the blockchain BEFORE public use. By establishing timestamped proof of ownership on Solana, Solturio creates an immutable chain of custody that protects against copycats, especially in the DeFi/DEX ecosystem where logo theft is rampant.

Prerequisites

- Node.js 18+
- PostgreSQL database (Neon serverless recommended)
- Pinata account for IPFS storage
- Solana RPC endpoint (devnet for testing, mainnet for production)

Environment Variables

Required:

DATABASE_URL, PINATA_API_KEY, PINATA_SECRET_KEY, PINATA_JWT, SESSION_SECRET, WALLET_ENCRYPTION_KEY

Optional:

TELEGRAM_BOT_TOKEN, SENDGRID_API_KEY, SC_API_URL, SC_API_SECRET

Installation

```
npm install
```

```
npm run db:push
```

```
npm run dev
```

The app will be available at <http://localhost:5000>

How to Use

1. Sign In - Click "Sign In" to authenticate using Replit Auth.
2. Create a Wallet - Navigate to Account page and create your .solturio.sol wallet
 - Standard wallet (e.g., 042.solturio.sol) - 0.1 SOL
 - Premium wallet (e.g., yourname.solturio.sol) - 0.15 SOL
3. Upload Logos - Drag & drop images or add URL links
4. Create Collections - Group related logos with metadata
5. Mint Collection - Generate verified images, upload to IPFS, create NFT certificate
6. View Collections - See NFT addresses, IPFS links, and verified previews

Key Features

Pre-Registration Workflow

- Register First, Use Everywhere - Upload logos before launching tokens
- IPFS Permanent Storage - Decentralized, immutable storage
- Blockchain Timestamps - Solana-based proof that predates copycats
- Gold Verification System - Verified logos display gold check badges

DEX Anti-Copycat Protection

- Real-time Verification API - DEX platforms verify logos in <100ms
- Automatic Copycat Detection - File hash comparison identifies stolen logos
- DMCA Automation - One-click takedown notices with blockchain evidence

Authorized Usage Tracking

- Pre-register official locations where logos will appear
- Platform-specific tracking across websites, social media, DEXs
- Verification records for IP disputes

Legal Documentation

- Prior Art Certificates - Blockchain-verified proof of first use
- DMCA Notices - Auto-generated with QR codes
- Cease & Desist Letters - Professional legal documents

IP Education Quiz

- Telegram Bot with daily quizzes on IP topics
- Game Points & Experience competitive scoring
- Leaderboards to track progress

Tech Stack

Frontend: React 18, TypeScript, Vite, TailwindCSS, shadcn/ui

Backend: Node.js, Express, TypeScript

Database: PostgreSQL (Neon serverless), Drizzle ORM

Blockchain: Solana, Metaplex SDK

Storage: IPFS (via Pinata), user-controlled .solturio.sol wallets

Authentication: Replit Auth (OpenID Connect)

Bot: Telegraf for Telegram quiz bot

Target Users

- Token Creators - Protect logos/tickers before token launch
- NFT Collections - Establish provenance for artwork and branding
- DEX Platforms - Integrate verification to protect users from scams
- Brand Owners - Document IP ownership with blockchain proof
- Artists & Designers - Register artwork with timestamped proof

Payment

Solturio runs on \$CATH (CATH Ecosystem):

- IP Registration: 100 \$CATH tokens
- Standard Wallet: 0.1 SOL
- Premium Wallet: 0.15 SOL

No fiat payments - crypto only (SOL, BONK, Arweave, \$CATH).

Security

- Encrypted Wallets - Private keys secured with AES-256-GCM
- Session Security - HTTP-only cookies, CSRF protection
- Wallet Restrictions - .solturio.sol wallets only hold certificates and SOL
- Key Handover Ceremony - Multi-stage secure key delivery process

License

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Solturio - Because in crypto, being first isn't enough. You need proof.